MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

Dept. of Information and Communication Technology

Class Test No. C1

4th Year 1st Semester, Session 20 21 to 22

Name: Riya Saha

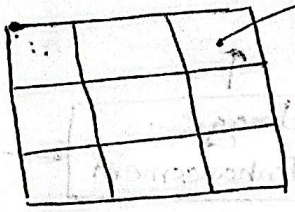
ID No: IT22049 Date:

Course Code: Course Title: DIP

14

Ans to the Question No-1

Digital image is type of image which made of many tiny small part called pixel.



Each pixel has bit K,

and the intensity level is 2^K

Digital image can be represented as matrices

Digital image can be function as $f(x,y)$

In general the origin is (0,0)

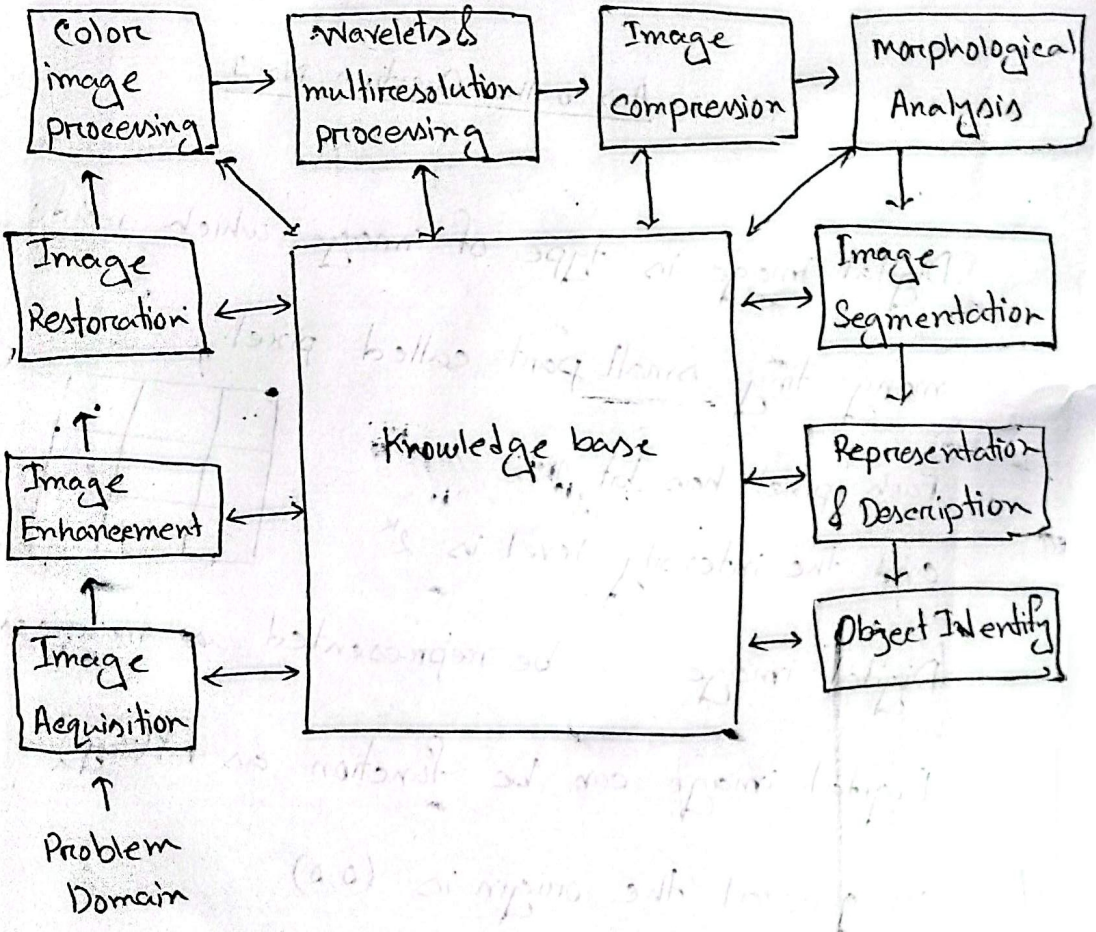
$x: 0 \rightarrow M-1$

$y: 0 \rightarrow N-1$

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

And do the Question No-2

The main steps involved in digital image processing



① **Image Aequisition:** In this part image was taken by any device. Example: Camera

② **Image Enhancement:** Image improves quality like

Image contrast, Image Sharpening

(3) Image Restoration: Reduce Image noise

(4) Color image processing: Improve image color information

(5) Wavelets and multiresolution: Improves the pixels of an image and image compression

(6) Image compression: Reduce size of image for low transmission cost

(7) Morphological Analysis: Improves image structure and shape

In this step the output image transfer to output attributes

(8) Image Segmentation: Segment the image into multipl

(9) Representation: It defines image in boundary or complete region.

Boundary is the external shape

region is the internal materials

(10) Object detection: Detecting the object into the image

Ans to the Question No-3

Sampling: Sampling is the process of converting continuous image into grid of pixel. Normally it is denoted into MxN where M is the rows and N is the columns

Quantization: Quantization is the process of converting continuous image ~~attributes~~ intensity values into discrete level. PGM is 256 → 4 levels

~~255~~

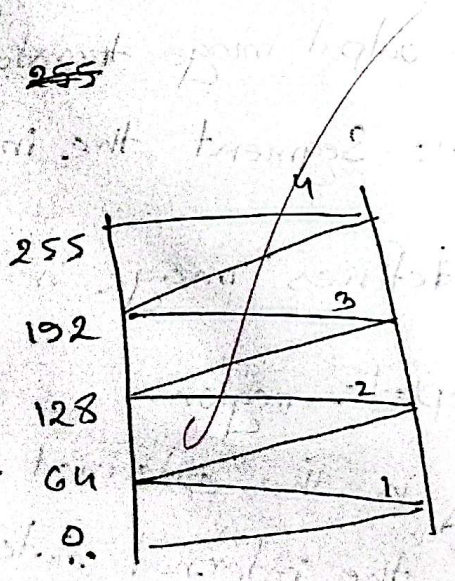


Fig: Quantization

In the process of image digitization sampling and Quantization is important.

After sampling and Quantization an image is transfe into matrices.

CamScanner

MAULANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

Dept. of Information and Communication Technology

Class Test No. 04

4th Year 1st Semester, Session 2021-2022

Name: Nahidur Rahman Tushar

ID No. IT22026 Date: 02.08.26

Course Code: ICT-4103 Course Title: Digital Image Processing

19

Ans: to the que: no: 01

MSE: MSE means Mean Square Error are used in digital image processing evaluation how different quality usage from the original image.

$$MSE = \frac{1}{mn} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y) - f'(x,y)]$$

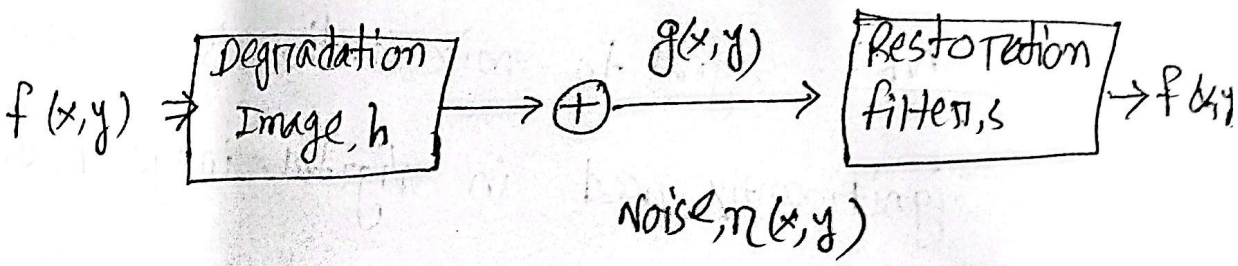
4

SNR: Signal to noise ratio is measured significantly used in digital image processing to evaluate the quality image from the original image.

$$SNR = \frac{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y)]^2}{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y) - f'(x,y)]^2}$$

Ans: to the que! no: 02

Image restoration: Image restoration is the process of continuing on reconstruct image of recovering noise blurs.



The degraded image in the spatial

represent is,

$$g(x,y) = h(x,y) * f(x,y) + \eta(x,y)$$

$h(x,y)$ is the spatial domain filter function.

The degraded image in the frequency domain filter is,

$$G(u,v) = H(u,v) * F(u,v) + \eta(u,v)$$

Ans: to the ques. no: 03

Explaining the six essential noise models in digital image processing :

① Gaussian noise :

Random signal that enters system to converge from digital signal .

The PDF is,

$$f(x,y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\bar{z})^2}{2\sigma^2}}$$

where,

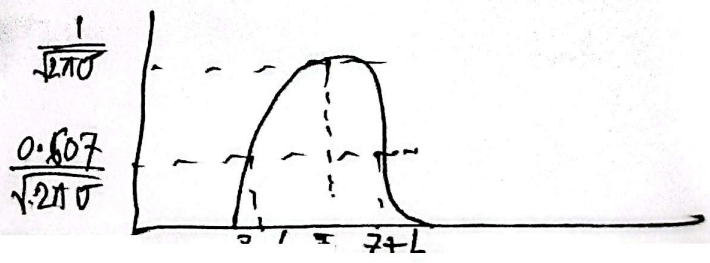
z = gray level

$\bar{z}$ = average mean value of z

σ = standard deviation

σ^2 = variance

figure:

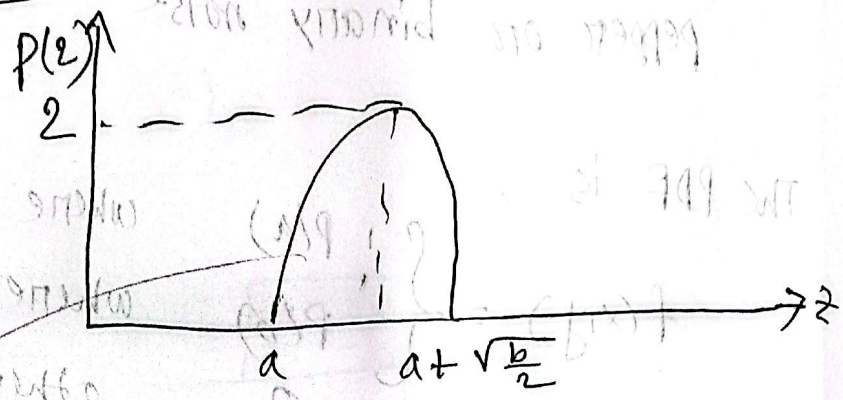


(ii) Rayleigh noise:

The PDF is,

$$f(x,y) = \begin{cases} \frac{2}{b}(z-a)\, e^{\frac{(z-a)^2}{b}} & \text{where } z \geq a \\ 0 & \text{otherwise} \end{cases}$$

figure:



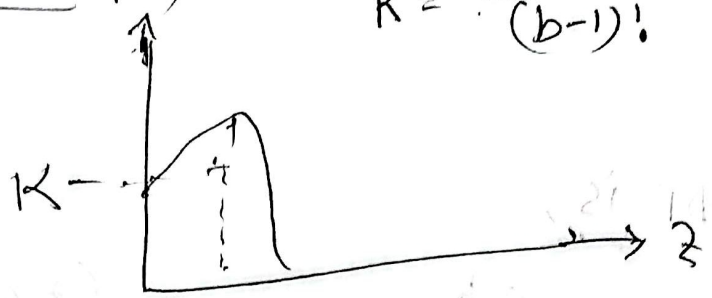
(iii) Gamma noise:

This types of noise mostly used or occurs in illumination problem.

The PDF is,

$$f(x,y) = \begin{cases} \frac{a^b (z)^{b-1}}{(b-1)!} e^{-(b-1)} & \text{where } z \geq a \\ 0 & \text{otherwise} \end{cases}$$

Figure: P(z)



$$K = \frac{a^b (z)^{b-1}}{(b-1)!} e^{-(b-1)}$$

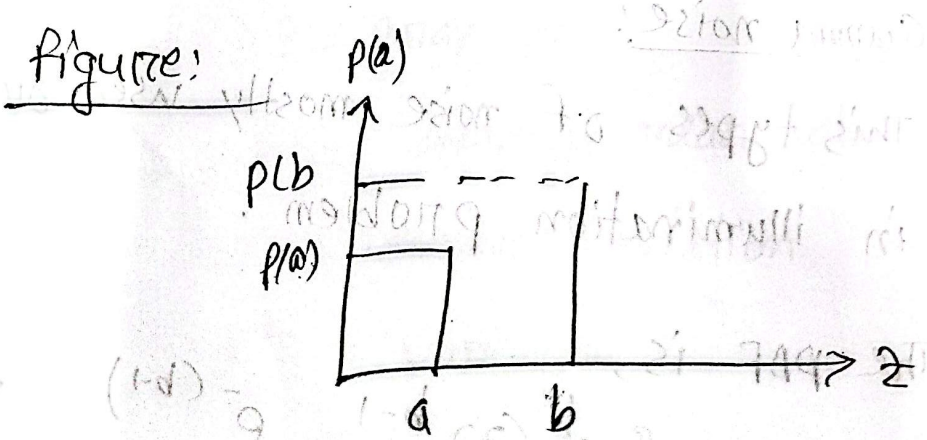
(iv) Impulse noise:

This type of noise also called salt and pepper or binary noise.

The PDF is

$$f(x,y) = \begin{cases} P(a) & \text{where } z = a \\ P(b) & \text{where } z = b \\ 0 & \text{otherwise} \end{cases}$$

Figure:



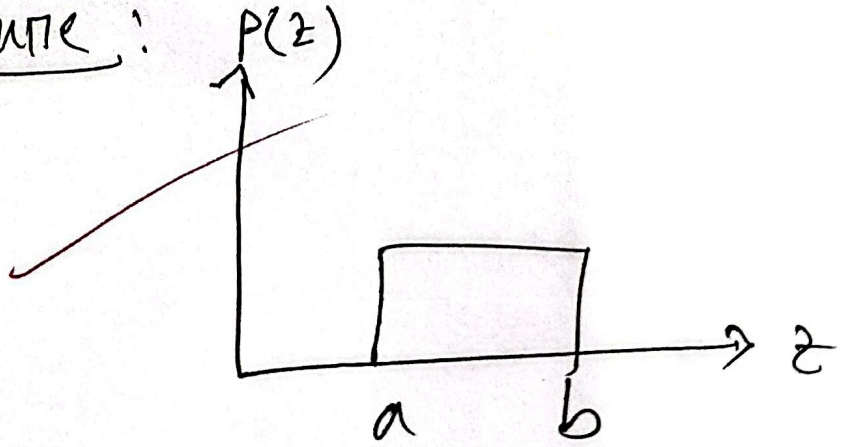
(v) uniform noise:

uniform noise is used in digital Image processing as polar impulse models.

The PDF is.

$$f(x,y) = \begin{cases} \frac{1}{b-a} & \text{where } a \le z \le b \\ 0 & \text{otherwise} \end{cases}$$

figure:



(vi) periodic noise.

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2| Image degradation and restoration process model:

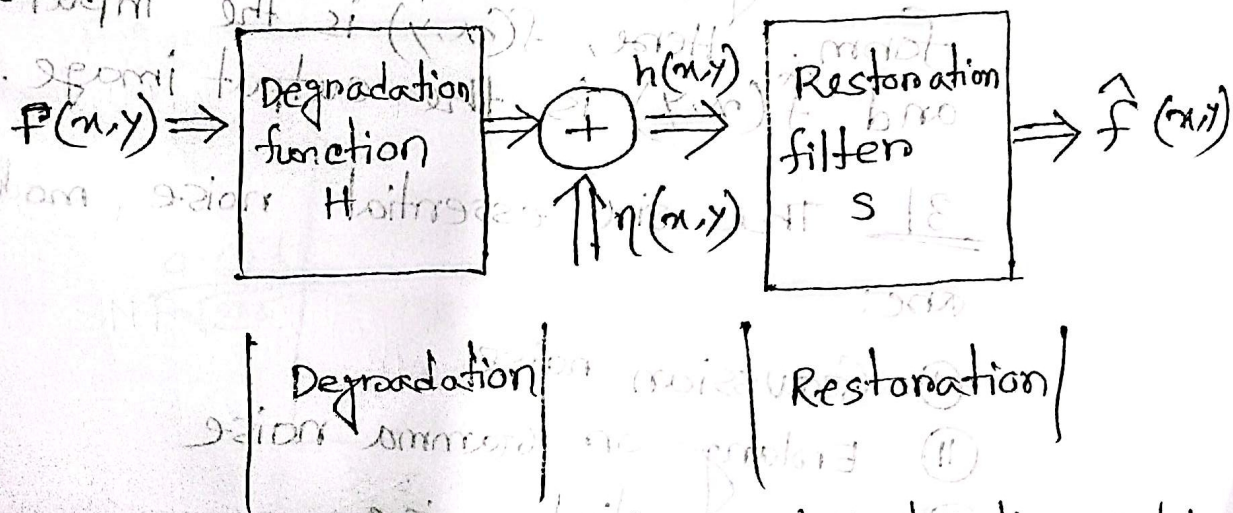


fig: degradation and restoration model.

Here, showing the objective process of restoring or recovering a degraded image to its original form or undistorted form. Here, $\eta(x,y)$ is the noise function and By restoration filter(s) we smooth the degraded image.

In that figure we can see that how the degradation of the image occurs by adding noise. Then by using the restoration filters we can again smoot smooth and restore the degraded image to its original form. Here, $f(x,y)$ is the input image and $\hat{f}(x,y)$ is the output image.

3| The six essential noise models are:

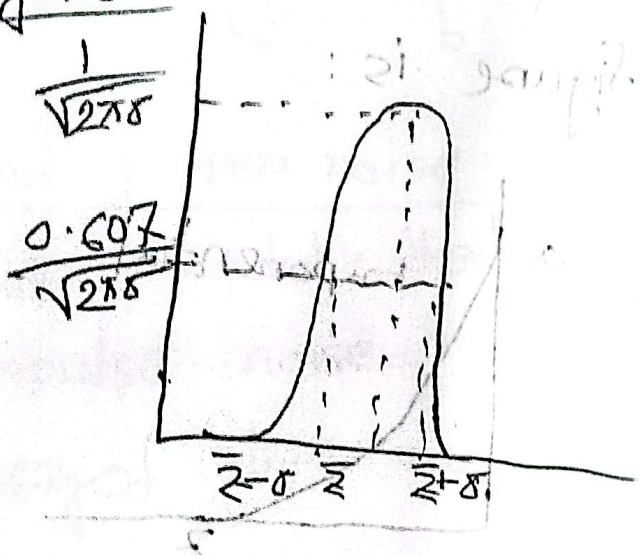
(I) Gaussian noise
(II) Erlang or Gamma noise
(III) Exponential noise
(IV) Uniform noise
(V) Rayleigh noise
(VI) Impulse or salt-and-pepper noise

Gaussian noise: It is most used noise model used for its simple mathematical calculation. This PDF is

$$P(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-\bar{z})^2/2\sigma^2}$$

here, $\bar{z}$ is the mean

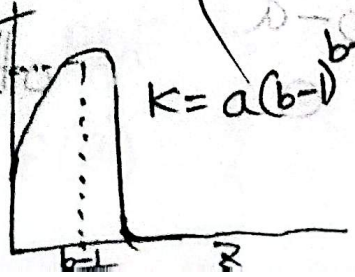
Figure:



Gamma noise: The PDF of this noise is expressed as:

$$P(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

figure:



$K = a(b-1)^{b-1} e^{-(b-1)}/(b-1)!$

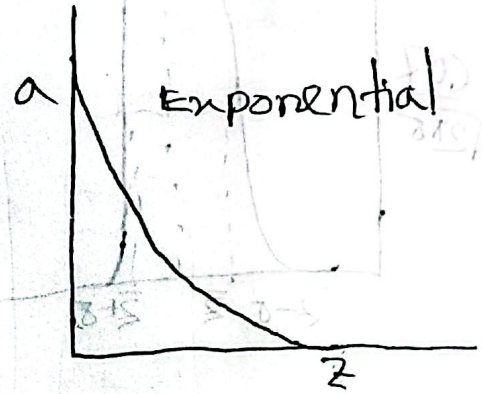
This is also best for skewed distribution.

Exponential noise: It is special type of gamma noise with $b=1$

PDF is expressed as:

$$P(z) = \begin{cases} ae^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

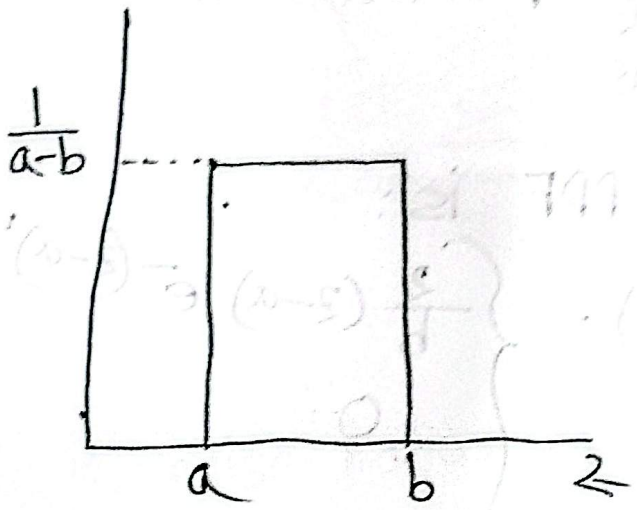
the figure is:



Uniform noise: The probability density function can be expressed as

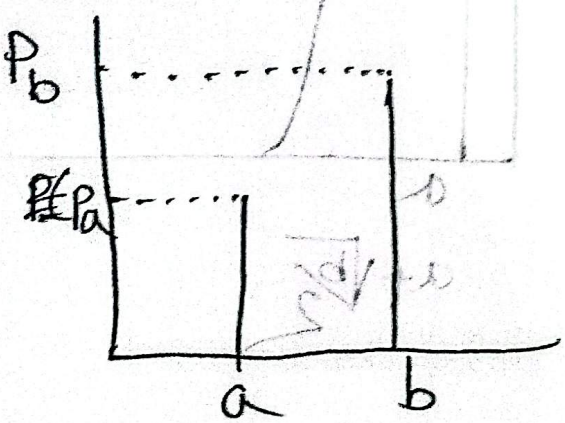
$$P(z) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$

the figure is:



Salt-and-pepper noise: It is the most used noise model, ~~that eliminate~~ It is also called impulse noise model.

The ~~PDF~~ fig of this model is:



and the PDF is:-

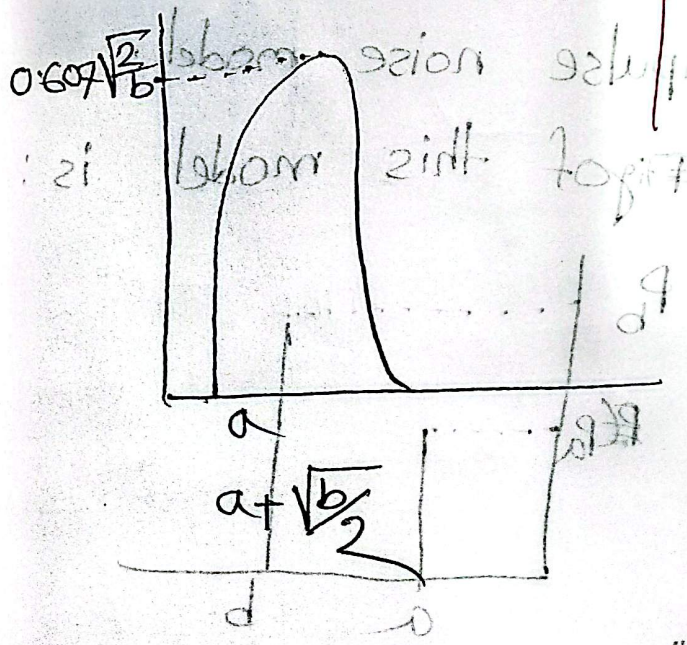
$$P(z) = \begin{cases} P_a & \text{for } z=a \\ P_b & \text{for } z=b \\ 0 & \text{otherwise} \end{cases}$$

Rayleigh noise: It is best for modelling positively skewed distribution functions.

The PDF is:

$$P(z) = \begin{cases} \frac{2}{b}(z-a)\, e^{-(z-a)^2/b} & \text{for } z \geq a \\ 0 & \text{for } z < a \end{cases}$$

Figure:



1) MSE: MSE stands for Mean square error.

$$MSE = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y) - \hat{f}(x,y)]^2$$

Or

Signal to noise ratio: It can be expressed as

$$SNR = 20 \log_{10} \frac{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y)]^2}{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y) - \hat{f}(x,y)]^2}$$